ABHILASH SAHOO

Associate Research Scientist,

Center for Computational Biology, &

Center for Computational Mathematics

Flatiron Institute, 162 5th Avenue NY, NY 10010

RESEARCH INTERESTS

- Development and Application of multi-scale minimalist physics and data based computational methods for understanding biophysical and biochemical systems.
- Cross-disciplinary applications of statistical data analysis methods, and network science.

EDUCATION

2016 – 2022 Ph.D. in Biophysics	University of Maryland, College Park Advisor: Dr. Silvina Matysiak
2011 – 2016 BS and MS in Physics	Indian Institute of Science Education and Research – Kolkata, India Advisor: Dr. Rumi De

WORK EXPERIENCE

Oct 2025 – Till now	**Associate Research Scientist,** *Structural and Molecular Biophysics, Flatiron* *Manager-* **Dr. Sonya Hanson** Multiscale modelling of microtubules, associated proteins, and intracellular transport
Jul 2022 – Oct 2025	**Flatiron Research Fellow,** *Structural and Molecular Biophysics, Flatiron Institute,* Manager- **Dr. Sonya Hanson** Multiscale modelling of microtubules, associated proteins, and intracellular transport.
Aug 2017 – Jul 2022	**Graduate Research associate**, *Department of Bioengineering,* *University of Maryland, College Park,* (Advisor- **Dr. Silvina Matysiak)** Assessment of physiological environment on neurodegenerative peptide aggregation
Aug 2016 – Dec 2016	**Teaching Assistant,** *Department of Physics, University of Maryland, College Park*
Aug 2015 – Dec 2015	**Teaching Assistant,** *Department of Physics***,** *Indian Institute of Science Education and Research Kolkata, India*
Aug 2015 – Aug 2016	**Research Assistant,** *Department of Physics***,** *Indian Institute of Science Education and Research Kolkata, India,* with **Dr. Rumi De** Theoretical Modelling of Collective Cell Migration
May 2014 – Aug 2014	**Visiting Summer Scholar**, *Institute of Mathematical Sciences, Chennai, India, with* **Dr. Vani Vemparala** Molecular Simulations of Block Polymers

JOURNAL PUBLICATIONS
(* represents equal contribution; $^{+}$ represents corresponding/senior author)

[1] **Abhilash Sahoo**, Hongcheng Xu, and Silvina Matysiak. Pathways of amyloid-beta absorption and aggregation in a membranous environment. Physical Chemistry Chemical Physics 21.16 (2019): 8559-8568

[2] **Abhilash Sahoo** and Silvina Matysiak. "Computational insights into lipid assisted peptide misfolding and aggregation in neurodegeneration" Physical Chemistry Chemical Physics 21 (41), 22679-22694

[3] **Abhilash Sahoo** and Silvina Matysiak. "A Microscopic Picture of Calcium-assisted Lipid Demixing and Membrane Remodeling using Multi-scale Simulations" The Journal of Physical Chemistry B 124 (34), 7327-7335

[4] **Abhilash Sahoo** and Silvina Matysiak. " Effects of Applied Surface-tension on Membrane-assisted Amyloid-beta Aggregation" Physical Chemistry Chemical Physics 23, 20627-20633

[5] **Abhilash Sahoo***, Pei-yin Lee* and Silvina Matysiak. " A Transferable, Explicit-Solvent Polarizable Coarse-Grained Model for Proteins" J. Chem. Theory Comput. 2022, 18, 8, 5046–5055

[6] Pei-yin Lee*, **Abhilash Sahoo*** and Silvina Matysiak., Folding and modulation of the helical conformation of Glycophorin A by point mutations. Physical Chemistry Chemical Physics. 2023, 25, 15, 10885-10893.

[7] Meenal Jain*, **Abhilash Sahoo***, and Silvina Matysiak. " Modulation of Amyloid-beta aggregation by Glucose." Physical Chemistry Chemical Physics 26, 6, 5038-5044

[8] Neha Nanajkar*, **Abhilash Sahoo***, and Silvina Matysiak. "Unraveling the Molecular Complexity of N Terminus Huntingtin Oligomers: Insights into Polymorphic Structures." J. Phys. Chem. B 2024, 128, 32, 7761–7769

[9] **Abhilash Sahoo**, and Sonya M. Hanson "Martini without the twist: Unveiling a mechanically correct microtubule through bottom-up coarse-graining in Martini 3 *PNAS Nexus* 4.7 (2025): pgaf202-pgaf202.

[10] Yikai Liu, Zongxin Yu, Richard J. Lindsay, Guang Lin, Ming Chen, **Abhilash Sahoo**$^{+}$ and Sonya M. Hanson^{+} "ExEnDiff: An Experiment-Guided Diffusion model for protein conformational ensemble generation." bioRxiv 2024.10.04.616517 [*accepted in PRX Life*]

[11] Eugene Formuichuk, **Abhilash Sahoo**, Camilla Edwards, Shrey Shah, Robert Oakes, Silvina Matysiak, and Christopher Jewell "Integrating Molecular Dynamics Simulations to Enable Rational Assembly of Immune Signals for Immunotherapy" (*submitted*)

[12] Neha Nanajkar*, **Abhilash Sahoo***, Shelli L. Frey, and Silvina Matysiak "Investigating curvature sensing by the Nt17 domain of Huntingtin protein" *ACS Chem. Neurosci.* 2025, 16, 17, 3282–3291

[13] Dewan, Sherwin, Ismail Shaikh, Connie Shaw, **Abhilash Sahoo**, Akshita Jha, and Alisha Pradhan. "Examining Age-Bias and Stereotypes of Aging in LLMs." *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*. 2025

[14] Mofidi, S. Mahsa, Abhilash Sahoo, Christopher J. Edelmaier, Stephen J. Klawa, Ronit Freeman, Amy Gladfelter, M. Gregory Forest, Ehssan Nazockdast, and Sonya M. Hanson. "Study of Protein-Protein Interactions in Septin Assembly: Multiple amphipathic helix domains cooperate in binding to the lipid membrane." *bioRxiv* (2025): 2025-08

TALKS

Biophysical Society Annual Meeting – 2022
Platform Presentation, San Francisco, US
Assessment of physiological environment on neurodegenerative peptide aggregation; and
A Transferable, Polarizable Coarse-grained Model for Simulations with Proteins.

American Chemical Society – 2021
Invited talk, San Francisco, US
A Transferable, Polarizable Coarse-grained Model for Simulations with Proteins.

American Physical Society – 2021
Platform Presentation, Boston, US
Effects of membrane curvature on amyloid-beta aggregation

University of Maryland, Bioengineering Department Invited Talks – 2019
College Park, US
Pathways of amyloid beta aggregations in membranous environments.

HONORS AND AWARDS

2021-2022	**Ann G. Wylie Dissertation Fellowship** University of Maryland
2021	**Biophysical Society Travel Award** Biophysical Society
2020	**Outstanding Graduate Research Assistant Award** University of Maryland
2019	**Jacob K. Goldhaber Travel Award** University of Maryland
2018-2022	**National Science Foundation – COMBINE Fellowship** University of Maryland
2016	**Dean's Fellowship** University of Maryland
2014	**Visiting Summer Research Fellowship – Physics** Institute of Mathematical Sciences (IMSc), Chennai, India
2011-2016	**Department of Science and Technology – INSPIRE Fellowship** Indian Institute of Science Education and Research - Kolkata
2011	**Qualified – Indian Institute of Technology Joint Entrance Examination** India, Selection rate: 0.2%

TEACHING, SERVICE & MENTORSHIPS

Paper reviewer
Journal of Chemical Theory and Computation (ACS), Journal of Physical Chemistry (ACS), Nature Communications, Proceedings of National Academy of Sciences, Communications Biology, Physical Chemistry Chemical Physics (RSC) and Research Advances (RSC).

Científico Latino Graduate Student Mentorship Initiative, 2023
Simons Foundation, NY
The program helps graduate school applicants from minoritized backgrounds by pairing them with STEM professionals in their respective STEM disciplines.

Center for Computational Biology Molecular Dynamics Course, 2024
Simons Foundation, NY
Designed and instructed a course on molecular dynamics.

Graduate Research Mentor – GRAD-MAP Winter workshop, 2019
University of Maryland, College Park

Workshop Organizer, Graduate research mentor and Python boot-camp instructor at winter workshop (2019) organized by Graduate Resources for Advancing Diversity with Maryland Astronomy and Physics (GRAD-MAP)

Graduate Teaching Assistant, 2017
University of Maryland, College Park
Primary instructor for undergraduate physics laboratory.

Graduate Teaching Assistant, 2016
Indian Institute of Science Education and Research Kolkata, India
Teaching assistant for a course in statistical mechanics